



Decision Support System for Sri Lankan Paddy Cultivation

Comments for the Proposal and how we addressed them

Comment from Panel	How we Addressed
what the logic to recommend suitable crops? How is the price trend predicted ?	Our system evaluates crop suitability by analyzing three key factors: the nitrogen, phosphorus, and potassium (N, P, K) levels in the soil, prevailing weather conditions and overall soil characteristics .Market price trends are predicted using the last three months of average crop prices.
What is the logic to recommend fertilzation and how many crop diseases are been detected?	Achieving real time data monitoring through IoT device supporting fertilization, crop recommendation and desease detection component
Previous “Growth stage-aware irrigation/fertilization” component overlaps with IT22178718 and is difficult to implement	Component fully changed to Disease-Aware Real-Time Paddy Harvest Yield Estimation
This component overlaps with IT22178718 component disease detection. How many crop diseases are detected?	IT22178718 component handles IoT-based soil and environmental monitoring. This component focuses on visual disease identification through leaf image analysis using YOLOv8 model. This component



Design Excellence of the Overall Solution

Context-aware IoT Field Monitoring



Disease-Aware Harvest Yield Estimation

Market-Aware Crop Recommendation

AI Disease Detection + Severity Analysis

Centralized React Dashboard with scenario-based financial insights

Problem & Design Excellence



The Problem & Its Impact/Gravity

- Paddy farmers lose 10–40%+ yield annually due to Rice Blast and Brown Spot.
- Over 1.8 million households face income instability.
- Existing systems are isolated — no real-time financial impact assessment of disease treatment.
- Cyclone Ditwah (2025) damaged over



Who Benefits?

- Small & Medium -scale paddy farmers
- Agriculture cooperatives
- Department of Agriculture officers
- Agri - insurance companies

Smart Crop Recommendation



Specialization : Information Technology

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How the Solution Addresses the Sub-Problem

- It provides simple and understandable suitability results, reducing the difficulty of manual crop selection and improving farming efficiency.

- This prototype provides crop recommendations and displays suitability levels in three stages
  - Highly Suitable
  - Moderately Suitable
  -Marginally Suitable

- As a result, farmers can improve overall agricultural productivity and profitability.

User Requirements Addressed

- Farmers grow only the crops that truly match their soil conditions and avoiding wasted resources on incompatible varieties.

- Farmers can prioritize the most profitable crop at the time of planting by analyzing the last three months of market price data

Design Excellence / Contribution

Crop type Selection -

The three-stage recommendation gives farmers a ranked choice -so they allowing farmers to clearly understand and compare recommendations

layered decision logic -

The system checks four factors step by step — first soil pH removes unsuitable crops, then nutrients find the best matches, weather confirms the right growing season, and finally market price selects the most profitable option.





Fertilization Decision Support with IoT monitoring Device Assistance

Specialization : Information Technology

How the Solution Addresses the Sub-Problem

- IoT device monitors soil properties and other factors required to proceed with **Fertilzation Component** & **Crop recommendation component**.

- Predictions made for 2 factors:
  - Fertilization Timing
  - Fertilization Quantity

- Random Forest Models used for considering fertilization timing.
- XGBoost regression for fertilization quanitity.

User Requirements Addressed

- System continously monitors growth stage, Soil moisture, pH values and npk values.
- Model outputs fertilization timing and quantity.
- Model accuracy has reached 90% for fertilization quantity.

Design Excellence & User Feedback

Design Excellence

- First soil data driven fertilization sytem tailored for paddy crops in Yala Season.
- Comparatively low cost:
  - IoT Device parts availability
  - No High-end devices required since it is a PWA

User Feedback

Positive: Ability to measure the fertilization amount per required area.

Negative: Nitrogen, Phosphorous, Pottasium requirements differ considering situations.





Disease-Aware Real-Time Paddy Harvest Yield Estimation

Specialization : Information Technology

How the Solution Addresses the Sub-Problem

- Takes real-time IoT sensor data and disease severity scores.
- Predicts expected harvest yield under three scenarios:
  - Healthy crop
  - Untreated crop
  - Treated crop
- Uses XGBoost + Linear Regression + Random Forest
- Outputs quantified financial impact (kg loss vs treatment cost).

User Requirements Addressed

- Know exact financial benefit before spending on treatment.
- Real-time, scenario-based yield & revenue forecasts.
- Actionable treatment recommendations with ROI.

Design Excellence / Contribution

- **Core Novelty**: First system to combine live IoT sensor data + disease severity with IRRI models to deliver financially quantified, scenario-based harvest predictions tailored for Sri Lankan paddy varieties (BG 300, AT 362).
- Seamless FastAPI integration with other group modules.
- Interactive React visualizations showing net treatment benefit.





Diseases Detection and Severity Estimation

Specialization : Information Technology

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How the Solution Addresses the Sub-Problem

- Sytsem identify the disease of the uploaded paddy leaf image and severity of the disease has spread.

- Predictions made for 3 outputs:
  - Disease Name (Rice Blast / Brown Spot)
  - Severity Category (Healthy-Severe)
  - Severity Score

- Uses YOLOv8 model to identify the diseased regions on the leaf and measures the spreading area.
- Based on the affected leaf area, system determines the severity of the disease.

User Requirements Addressed

- Farmer uploads a diseased paddy leaf image.
- Know how severely the disease has spread across the leaf area.
- Receive disease name and severity instantly.

Design Excellence / Contribution

- **Core Novelty**: First system localized for Sri Lankan paddy diseases with severity based on leaf area spread.
- Farmers can now understand how far the disease has spread on the selected paddy crop.
- Trained exclusively on locally collected Sri Lankan field images.

Commercialization



Commercialization / Sustainability of Solution

Customer Persona

Smallholder paddy farmers (25–55 years), tech-adopting younger farmers, and farmer cooperatives in major paddy districts.

Market Size

1.8 million+ paddy farming households in Sri Lanka.

What is Unique

Real-time disease-aware scenario-based yield & revenue forecasting integrated with IoT, severity analysis, and market intelligence — fully localized for Sri Lankan conditions.

How Cost will be Recovered

- Freemium model (Basic free, Premium LKR 500–1,500/month)
- Cooperative & B2B subscriptions
- API licensing for insurance companies & government

Thank You!

